12.7.1 Transformation

Transformation is genetic alteration of a cell resulting from the direct uptake, incorporation and expression of exogenous genetic material (exogenous DNA) from its surroundings. Transformation occurs naturally in some species of bacteria, but it can also take place by artificial means in other cells. For transformation to happen, bacteria must be in a state of competence. Competence refers to the state of being able to take up exogenous DNA from the environment. There are two forms of competence: natural and artificial. Transformation works best with DNA from closely-related species. The naturally-competent bacteria carry sets of genes that provide the protein machinery to bring DNA across the cell membrane(s).

There are some differences in the mechanisms of DNA uptake by gram positive and gram negative cells. However, they share some common features that involve related proteins. The DNA first binds to the surface of the competent cells on a DNA receptor, and passes through the cytoplasmic membrane via DNA translocase. Only single stranded DNA may pass through, one strand is therefore degraded by nucleases in the process, and the translocated single-stranded DNA may then be integrated into the bacterial chromosomes. Figure 12.16 shows mechanism of transformation.

Artificial competence can be induced in laboratory procedures that involve making the cell passively permeable to DNA. Typically, the cells are incubated in a solution containing divalent cations; most commonly, calcium chloride solution under cold condition, which is then exposed to a pulse of heat shock. Electroporation is another method of promoting competence. Using this method, the cells are briefly shocked with an electric field of 10–20 kV/cm which is thought to create holes in the cell membrane through which the plasmid DNA may enter. After the electric shock, the holes are rapidly closed by the cell's membrane-repair mechanisms.

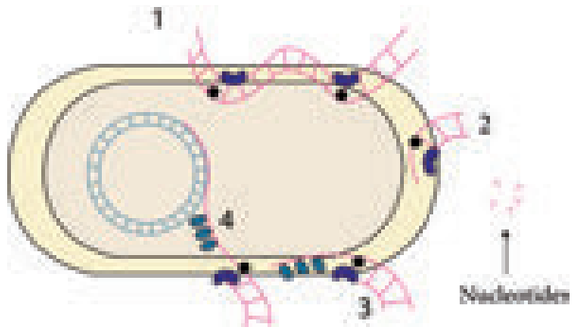


Figure 12.16: Mechanism of transformation **1.** Binding of DNA; **2.** Degradation of one strand; **3.** Entry of ssDNA; **4.** Integration into host chromosome.

12.7.2 Conjugation

The initial evidence for bacterial conjugation, came from an experiment performed by Joshua Lederberg and Edward L Tatum in 1946. Later in 1950, Bernard Davis gave evidence that physical contact of the cells was necessary for conjugation. During conjugation, two live bacteria (a donor and a recipient) come together, join by cytoplasmic bridges (e.g. pilus) and transfer single stranded DNA (from donor to recipient).

Inside the recipient cell, the new DNA may integrate with the chromosome (rather rare) or may remain free (as is the case with plasmids). Conjugation can occur among the cells from different genera of bacteria, while transformation takes place among the cells of a bacterial genus.

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